

# Current Affairs

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# Chapter 1

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## Polity

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### 1.1 Indian Constitution, Citizenship, Union and its Territory, FR, DPSP, FD

### 1.2 Executive, Legislature & Judiciary

### 1.3 Local Government, Constitutional Bodies, non-constitutional Bodies, Election

### 1.4 governance

### 1.5 bills

#### 1.5.1 Transition from MGNREGA to VB-GRAM G Act, 2025: Analysis of Draft Rules

##### • Structural Changes: VB-GRAM G vs. MGNREGA:

1. **Employment Days:** Increases guaranteed rural wage employment from 100 days to **125 days** per year.
2. **Allocation Model:** Flips the bottom-up, demand-driven model of MGNREGA (based on state labor budgets) to a top-down “**Normative Allocation**” system determined by the Centre.
3. **Funding Devolution Formula:** Center will use the **16th Finance Commission’s** horizontal devolution recommendations to determine state-wise normative allocation.
4. **Fiscal Burden:** Moves away from the Center paying 100% of the wage bill under MGNREGA; shifts a significant portion of the funding burden onto states. States must bear 100% of any expenditure incurred in excess of their normative allocation.
5. **Agricultural Seasonal Clause:** Introduces a mandatory **60-day pause** during peak sowing and harvesting seasons to prevent rural farm labor shortages.
6. **Performance-Linked Allocations:** A portion of the normative fund will be kept aside and distributed based on state efficiency indicators: timely wage payments, compliance with social audits, and percentage of project completion within a financial year.

##### • Worker Transition Framework:

- **Card Validity:** Existing MGNREGA job cards remain valid for seeking employment under VB-GRAM G, provided they are renewed and verified through e-KYC. This acts as a transitional arrangement until state governments issue new **Gramin Rozgar Guarantee Cards**.
- **Continuity of Legal Guarantees:** Both the **Unemployment Allowance** and the **Direct Benefit Transfer (DBT)** mechanism into verified bank or post office accounts are retained.
- **Pending Clause:** The Union government is yet to declare the minimum wage rate under the new scheme.

##### • Critical Analysis & Concerns:

- **Dilution of Rights-Based Approach:** Shifting from a demand-driven model to a pre-fixed central normative allocation risks weakening the legal “right to work” tenet that formed the core of MGNREGA.
- **Implications for Fiscal Federalism:** Forcing states to bear excess expenditure beyond the top-down normative limit, combined with decreased allocations for historically proactive states, could trigger severe Centre–State fiscal friction.
- **Rural Wage Dynamics:** While the 60-day seasonal pause protects landowners by maintaining farm labor availability, it may restrict immediate alternative income generation for vulnerable landless laborers during transition windows.

#### 1.5.2 Promotion and Regulation of Online Gaming (PROG) Rules, 2026

##### • Sectoral Classification & Market Data:

- **Market Scale:** Indian online gaming market valued at INR 232 billion (2024), projected to grow at 11% CAGR to INR 316 billion by 2027.

Transaction-based gaming constitutes 77% of total revenue.

- **Esports:** Competitive, strategy, and skill-based digital sports integrated into multi-sport events.
- **Online Social Games:** Casual, skill-based entertainment or learning games (classified as safe).
- **Online Money Games:** Platforms involving financial stakes (chance/skill mix). Impacted ~45 crore people with cumulative losses exceeding Rs. 20,000 crores.

#### • Core Pillars of the Framework:

- **Online Gaming Authority of India:** Digital-first attached office under MeitY, chaired by Additional Secretary (MeitY) with JS-level members from Home, Finance, I&B, Sports, and Law.
- **Determination Mechanism:** Objective evaluation (stakes, monetary winnings, asset monetization) completed within 90 days via suo motu action or application to classify games.
- **Registration Regime:** Mandatory for e-sports and notified social games. Issues a digital Certificate of Registration valid up to 10 years. Money games are explicitly excluded from registration.
- **User Safety Mandates:** Technical and behavioral safeguards including age-gating, time limits, parental controls, data retention, and integrity monitoring.

#### • Key Issues Addressed:

- **Social Risks:** Rising incidence of digital addiction, severe financial distress, money laundering, and related suicides.
- **Regulatory Ambiguity:** Overlapping definitions of chance versus skill that previously enabled illegal betting platforms to masquerade as casual games.

#### • Enforcement and Penalties:

- **Prohibitions:** Complete ban on offering, advertising, or facilitating online money games.

Banks/payment gateways barred from processing related transactions.

- **Criminal Liabilities:** First offense for operating money games attracts up to 3 years imprisonment and/or Rs. 1 crore fine. Repeat offenses carry 3–5 years imprisonment and Rs. 1–2 crore fine. Illegal advertising attracts up to 2 years imprisonment or Rs. 50 lakh fine.
- **Investigating Agency:** Empowered Cyber Cell officers at State/UT police station, district, and Commissionerate levels. Digital blocking executed via the IT Act, 2000.

#### • Way Forward:

- Harmonize state-level gambling laws with the federal PROG Act to avoid jurisdictional conflicts.
- Establish robust algorithmic filters to dynamically track and block proxy mirror sites of banned money games.
- Boost indigenous game development ecosystem leveraging the explicit legal safety provided to e-sports.

## 1.6 judgements

### 1.6.1 Supreme Court Judgement on Special Intensive Revision (SIR) of Electoral Rolls

#### • Four Core Legal Issues Framed by the Court:

1. Whether the ECI possesses the structural power to conduct the impugned SIR.
2. Whether the SIR is founded on a legitimate purpose, and whether the adopted measures are proportionate to the object sought to be achieved.
3. Whether the procedure adopted is contrary to, or in violation of, the provisions of the *Representation of the People Act, 1950* and the *Registration of Electors Rules, 1960*.
4. Whether, under its constitutional mandate to prepare and maintain ERs, the ECI is empowered to scrutinize the citizenship status of individuals seeking inclusion or continuation in the rolls.

- **Key Supreme Court Observations & Remarks:**

1. **Intersection of Constitutional Concerns:**

The SC noted that the issue rests at the intersection of two constitutional duties: including all eligible citizens, and ensuring that the ER reflects the true composition of the political community.

2. **Institutional Deference:** The Court delivered a complete "clean chit" to the ECI, characterizing its process and methodology as "**optimal**". It observed that it is not open to the Court to supplant its own judgment in matters concerning execution for which the ECI is endowed with specialized "**institutional expertise**".

3. **Systemic Nature of Errors:** The Court was convinced that cumulative inaccuracies over an extended period had become systemic in nature. It recognized that such deep-seated inaccuracies require a comprehensive SIR because they do not readily admit to piecemeal or localized solutions.

4. **Limited Inquiry vs. Citizenship Finality:** The SC explicitly reaffirmed that while the ECI is empowered to undertake a limited inquiry into citizenship to satisfy voter eligibility, this does not equal a final legal determination of citizenship in a strict sense. It merely reflects the ECI's contextual inability to be satisfied for electoral inclusion, leaving the voter's ultimate claim to citizenship open for competent statutory adjudication.

5. **Test of Proportionality:** The Court held that the exercise successfully met the parameters of proportionality, as it integrated sufficient procedural safeguards to prevent arbitrary or widespread disenfranchisement. Post-exercise data did not demonstrate any systemic constitutional infirmity in design.

- **Directions Issued to the Election Commission of India (ECI):**

1. **Mandatory Reference Window:** The ECI is legally bound to disclose the exact count of "ineligible" persons discovered through the SIR and refer their names to the com-

petent authority under the *Citizenship Act, 1955* within a strict **four-week** deadline for formal citizenship adjudication.

2. **Strict Pre-election Timeline:** This adjudication must be completed by the competent authority, following due notice and hearings, **before** the next impending Parliamentary, Legislative Assembly, or Local Body elections in the respective state or constituency.

3. **Accountability and Public Disclosure:** Having received judicial backing, the ECI is now expected to present a complete, transparent report card comparing the post-SIR health of the electoral rolls (on parameters of integrity, accuracy, and purity) against prior rolls.

- **Critical Concerns and "Structural Correctives":**

1. **The Evolution of Safeguards:** The SC admitted that the SIR process, as initially designed, did raise legitimate concerns regarding documentation, transparency, and access.

2. **Judicial Infusion of Fairness:** The Court highlighted that its own progressive, interim interventions served as vital structural correctives. These interventions forced the ECI to adopt more inclusive standards, such as:

- (a) Treating the **Aadhaar card** as an additional, valid 12th document of identity verification.

- (b) Mandating the public disclosure of the draft roll containing approximately **65 lakh excluded electors** in Bihar, accompanied explicitly by the individual reasons for their deletion, ensuring a verifiable and contestable administrative process.

3. **The Scale of Deletions vs. Access to Justice:** Critical observers (including former election commissioners) note that while the court order delivers a massive legal victory to the ECI, massive hurdles remain for the affected populace. Hundreds of thousands of deleted individuals—such as the massive block of over **27 lakh electors excluded in**

**West Bengal**—face immense logistical hardships while navigating the appellate process and waiting for structural justice.

## **1.7 miscellaneous**



## Chapter 2

### Economy

#### 2.1 Pillar 1: Money, Banking, Finance & Insurance

#### 2.2 Pillar 2: Budget, Taxation & Public Finance

#### 2.3 Pillar 3: Balance of Payments (BoP) & International Trade

#### 2.4 Pillar 4: Sectors of Economy, GDP & Inflation

##### 2.4.1 Union Budget 2026–27: Key Manufacturing Hub Announcements

- **Chemicals:** 3 new **Chemical Parks** – cluster-based manufacturing; shared utilities, safety infrastructure, logistics connectivity.
- **Textiles:** Continued rollout of 7 **PM MITRA Parks** (Mega Integrated Textile Region and Apparel) – large-scale integrated textile manufacturing hubs.
- **MSME Manufacturing Clusters:** Strengthening cluster-based manufacturing + **Common Facility Centres (CFCs)** – shared production, testing, quality infrastructure.
- **Capital Goods / Industrial Support:** Support for **tool rooms, testing & calibration facilities** – boost domestic capital goods manufacturing capability.
- **Infrastructure (PM GatiShakti):** Continued multimodal, coordinated infrastructure planning linked to industrial/manufacturing locations.
- **Biopharma SHAKTI** (Global Biopharma & Manufacturing Hub):
  - Outlay: ₹10,000 crore / 5 years.
  - 3 new **NIPERs** (National Institutes of Pharmaceutical Education and Research).
  - **1,000+ accredited clinical trial sites** (nationwide network).
  - Objective: Position India as global biopharma manufacturing hub.

#### 2.5 Pillar 5: Infrastructure

##### 2.5.1 The Great Nicobar Project: Strategic, Economic and Ecological Framework

###### • Context & Strategic Significance:

1. Aimed at transforming Great Nicobar into a strategic maritime and economic hub in the Indian Ocean Region, strengthening India's presence in the Andaman Sea and Southeast Asia.
2. Seeks to capture transshipment trade currently lost to foreign ports like Colombo, Singapore, and Klang due to India's lack of deep-water berths.

###### • Core Infrastructure Components:

1. **International Container Transshipment Terminal (ICTT):** Located at Galathea Bay with a capacity of 14.2 million Twenty-Foot Equivalent Units (TEU).
2. **Greenfield International Airport:** Designed to handle 4,000 Peak Hour Passengers (PHP), projecting an initial 1 million passengers annually, scaling up to 10 million.
3. **Township and Area Development:** A planned urban center spanning 16,610 hectares to support port-led residential, commercial, and institutional requirements.
4. **Power Plant:** A 450 MVA hybrid gas and solar-based power plant designed with N-1 contingency (redundancy) to phase out diesel generators and reduce emissions.

###### • Forest Diversion & Compensatory Afforestation:

1. Diverts 1.82% of the total forest cover of the A&N Islands.
2. Preservation of 65.99 sq. km of land as strict green zones within the project area.

###### • Tribal Welfare & Rights Protection:

1. Inhabited by Particular Vulnerable Tribal

Groups (PVTGs): Shompens ( $\approx 237$ , hunter-gatherers) and Nicobarese ( $\approx 1,094$ , coastal fishing communities).

2. **No Displacement/Relocation:** Project aligns with the Shompen Policy (2015), Jarawa Policy (2004), and Article 338A(9) of the Constitution. No tribal habitations are being relocated.
3. **Tribal Reserve Land Balance:** 84.10 sq. km overlaps with the current Tribal Reserve. Within this, 11.032 sq. km is already utilized as revenue land since 1972. Thus, 73.07 sq. km is being de-notified for the project. To balance this, 76.98 sq. km is being re-notified as Tribal Reserve, creating a **net addition of 3.912 sq. km** to the reserve area.

## 2.6 Pillar 6: Human Resource Development (HRD)

### 2.6.1 Skilling India for a Future-Ready Workforce (Budget 2026–27 & Policy Framework)

#### Context & Macro Indicators

- Economic Survey 2025–26: employment-focused skilling bridges skill gaps, improves productivity, promotes social mobility.
- PLFS (Feb 2026): LFPR (15+ yrs) = **55.9%**; rising female participation; declining unemployment (rural + urban).
- Union Budget 2026–27: skill development as **cross-sectoral priority**.

#### Budget 2026–27: Sector-wise Skilling Initiatives

**Education** Allocation: **₹1.39 lakh crore** ( $\uparrow 8.27\%$  over BE 2025–26).

- **5 University Townships** near industrial/logistics corridors (universities, research, skill centres).
- **1 girls' hostel per district** via VGF/capital support (STEM retention).
- Upgradation/establishment of **4 telescope facilities**: National Large Solar Telescope, National Large Optical Infrared Telescope, Himalayan Chandra Telescope, COSMOS-2 Planetarium.

**Textiles & Apparel** Sector:  $\sim 2\%$  GDP, 11% manufacturing GVA, 8.63% exports; 45 million direct jobs; 6th

largest global exporter.

- **National Fibre Scheme** – self-reliance in natural/man-made/new-age fibres.
- **Textile Expansion and Employment Scheme** – modernise clusters.
- **National Handloom and Handicraft Programme** – integrate schemes, support weavers/artisans.
- **Tex-Eco Initiative** – sustainable/globally competitive textiles.
- **Samarth 2.0** – industry-academia textile skilling.
- **Mega Textile Parks** – scale + value addition in technical textiles.
- **Mahatma Gandhi Gram Swaraj Initiative** – khadi/handloom/handicrafts branding + skilling.

**Health** Outlay: **₹980 crore** (3 yrs) for allied/healthcare workforce.

- **Biopharma SHAKTI**: ₹10,000 crore/5 yrs; 3 new NIPERs + upgradation of 7 existing NIPERs; 1,000+ clinical trial sites; strengthening CDSCO.
- Upgradation of **Allied Health Professionals (AHP) institutions**; 10 disciplines (optometry, radiology, anaesthesia, OT Tech, applied psychology etc.); **1 lakh AHPs added in 5 yrs**.
- **Care Ecosystem**: NSQF-aligned programmes; **1.5 lakh caregivers** trained in FY2026–27.
- **5 Regional Medical Hubs** (State-private partnership): AYUSH centres, Medical Value Tourism facilitation, diagnostics, rehab.

**AYUSH** Allocation: **₹4,408 crore**.

- 3 new **All India Institutes of Ayurveda (AIIA)**.
- Upgradation of AYUSH pharmacies and Drug Testing Laboratories.
- Upgradation of **WHO Global Traditional Medicine Centre**, Jamnagar.

**Livestock** **₹6,153 crore**; contributes  $\sim 16\%$  to farm income.

- **Loan-linked capital subsidy:** private veterinary/para-vet colleges, hospitals, diagnostic labs; scale up veterinary professionals by **20,000+**.

**Orange Economy (AVGC)** Media & entertainment: ₹2.5 trillion (2024) → ₹3.06 trillion by 2027 (CAGR ~7%).

- **AVGC Content Creator Labs:** 15,000 secondary schools + 500 colleges.
- AVGC sector demand: **2 million professionals by 2030.**

### Design

- New **National Institute of Design** in eastern India.

**Tourism** Sector: 5.22% GDP (FY2024); **8.46 crore jobs** (13.3% total employment).

- **National Institute of Hospitality** – upgrade of National Council for Hotel Management & Catering Technology (NCHMCT); academia-industry-government bridge.
- Pilot: **upskilling 10,000 tourist guides** at 20 iconic sites (12-week hybrid course; IIM collaboration).
- **National Destination Digital Knowledge Grid** – digital documentation of cultural/heritage sites.

**Sports** Ministry allocation: ₹3,346 cr (RE 2025–26) → ₹4,479.88 cr (BE 2026–27) (↑ ₹1,133 crore).

- Target: **Top 10 by 2036; Top 5 by 2047.**
- **Khelo India Mission:** integrated talent pathway (foundational → elite), coach development, sports science integration, leagues/competitions, infrastructure.

*Flagship Skilling Schemes (Skill India Mission, launched 2015)*

### PMKVY / PMKVY 4.0

- Phases I–IV: pilot incentive → large-scale demand-driven skilling.
- **27.24 lakh trained** (38 sectors, 36 states, 737 districts) as of March 2026.

- April 2024–March 2026: **10.91 lakh trained** (ITeS, aerospace, agri, rubber, leather etc.).
- **69 customised courses + 154 future-skill job roles** (AI, Industry 4.0, green jobs, digital).
- **16,900+ institutions;** 6,800+ Skill Hubs in schools/HEIs/ITIs.

**Jan Shikshan Sansthan (JSS)** Origin: Shramik Vidyapeeth, 1967; 100% GoI grant via NGOs.

- Target: non-literates, neo-literates, school dropouts.
- **36,48,692 beneficiaries** trained (2018–March 2026); 26,519 tribal beneficiaries.
- 32 new NCVET-approved courses (NSQF Levels 3, 3.5, 4).
- Products marketed on **UdyamKart portal** (from Dec 2024).

**NAPS / NAPS 2.0** Launched August 2016; “earn while you learn” model.

- GoI contributes **25% stipend** (≤ ₹1,500/month) via DBT.
- **54.41 lakh+ apprentices** engaged (2016–March 2026).
- FY2025–26: **12.35 lakh engaged;** 6.42 lakh completed OJT.
- **Certificate of Proficiency (CoP):** launched Sep 2025; 67,711 CoPs generated by March 2026.
- FY2025–26 DBT: **₹562.75 crore** via 40.10 lakh transactions.

**Craftsmen Training Scheme (CTS)** Introduced **1950.**

- Training in **169 courses** via **14,688 ITIs** (3,345 Govt + 11,343 Pvt).
- 14 new CTS courses + 22 revised (3 years); enrolment ↑ 12.51 lakh (FY23) → **14.70 lakh (FY26).**

*PM–SETU (Pradhan Mantri Skilling & Employability Transformation through Upgraded ITIs)*

- Launched: **October 2025;** Centrally Sponsored Scheme; outlay: **₹60,000 crore.**

- **Hub-and-spoke model:** 1,000 Govt ITIs (200 hub + 800 spoke).
- **Special Purpose Vehicles (SPVs)** with Anchor Industry Partners (AIPs) for co-ownership/co-management.
- **5 National Skill Training Institutes** (Bhubaneswar, Chennai, Hyderabad, Kanpur, Ludhiana) → National Centres of Excellence for Skilling (global partnerships).
- States co-creating upgradation: Haryana, Karnataka, Tamil Nadu.

#### *Key Targets & Vision*

- **Viksit Bharat @2047:** human capital as core growth driver.
- Demand-driven training, measurable outcomes, inclusive access.
- Formalisation + productivity gains via healthcare, AVGC, tourism, care economy skilling.

## Chapter 3

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### Environment

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#### 3.1 Environmental Laws and policies

#### 3.2 Climate Change, Carbon Credit, ETS

#### 3.3 Environmental Convention and bodies

#### 3.4 Pollution

#### 3.5 Biodiversity

#### 3.6 Protected Areas

#### 3.7 Species

##### 3.7.1 Rediscovery of *Gynacantha khasiaca* (Long-tailed Duskhawker)

- **Context & Rediscovery:**

- A rare dragonfly species, *Gynacantha khasiaca* (commonly known as the long-tailed duskhawker), has resurfaced in the Changlang district of Arunachal Pradesh (**Namdapha National Park and Tiger Reserve**).

- **About Dragonflies:**

1. Possesses two large compound eyes, each comprised of thousands of tiny lenses and photoreceptor clusters.
2. Features a near-360° field of vision and possesses the distinct behavioral trait of being able to hover perfectly still in mid-air.
3. They serve as critical bioindicators and components of freshwater ecosystems, functioning simultaneously as apex invertebrate predators and vital prey within the aquatic food web.
4. long-tailed duskhawker has also been historically documented across Assam, Meghalaya, West Bengal, Uttarakhand, and Maharashtra.

#### 3.8 Miscellaneous

## Chapter 4

### International Relations

#### 4.1 Groupings

#### 4.2 Major Initiatives and Agreements

##### 4.2.1 Pax-Silica Initiative (US-led)

- **Led by:** US Department of State
- **Objective:** Secure a resilient, trusted supply chain for ACES (AI, Critical minerals, Electronics & Semiconductors).
- **India's Involvement:** India signed in during the India AI Impact Summit.
- **Latest Member:** Sweden
- **Member Nations:** Australia, Japan, ROK (Republic of Korea), UK, Singapore, Qatar, UAE, Greece, Israel, India, and Sweden.
- **Total Members:** 11

#### 4.3 Bilateral Relations

#### 4.4 India and the World

##### 4.4.1 India's West Asia Policy: Framework, Interventions, and Strategic Balancing

- **Core Pillars of Strategic Engagement:**
  1. **Energy Security:** Heavy reliance on regional crude oil and gas imports.
  2. **Human Security & Economy:** Protection of the massive Indian diaspora, safeguarding remittance inflows, and attracting sovereign wealth investments.
  3. **Geopolitics & Trade:** Critical for maritime shipping lanes, regional connectivity nodes, and defense/counter-terrorism partnerships.
- **The "Rubik's Cube" Diplomacy (Tri-Directional Balancing):**
  1. **Israel Pillar:** Anchored in core national security, defense procurement, advanced technology transfers, intelligence sharing, and counter-terrorism cooperation.
  2. **Gulf States Pillar (Saudi Arabia, UAE):** Vital economic, investment, energy, and food

security partners. Requires handling Gulf countries as distinct actors rather than a single homogeneous bloc.

3. **Iran Pillar:** Critical for continental connectivity, developing the strategic **Chabahar Port**, and maintaining alternative transit access to Afghanistan and Central Asia.

- **Emerging Regional Vulnerabilities & Security Interconnections:**

1. **Cascading Geopolitical Crises:** Interconnected flashpoints including the Gaza conflict, US-Iran tensions, and multi-front militia activities (e.g., in Lebanon).
2. **Maritime Security Risks:** Direct threats to trade and shipping routes in the Red Sea and Arabian Sea, causing cascading global impacts on supply lines.

- **Direct Macroeconomic and Social Impact on India:**

1. **Fiscal Strains:** Regional volatility directly triggers spikes in imported oil and fertilizer prices, depleting foreign exchange reserves and aggravating exchange-rate volatility.
2. **Expatriate Welfare:** Instability creates physical and economic security anxieties among the Indian expatriate worker communities.
3. **Commercial Disruptions:** Trade bottlenecks and maritime commerce delays directly affect India's export-import schedules.

- **Strategic Autonomy & Foreign Policy Way Forward:**

1. **De-hyphenated Engagement:** Rejection of exclusive alignment with either the Israeli or Arab blocs; absolute prioritization of multi-aligned strategic autonomy.
2. **Dialogue Maintenance:** Preserving multi-stakeholder diplomatic access while systematically avoiding strategic overcommitment

to any single regional sub-rivalry.

3. **"Access over Visibility" Doctrine:** Emphasizing high-yield backchannel diplomacy, quiet engagement, and continuous communication lines over public positioning or highly visible, polarising declarations during active conflicts.

#### 4.4.2 India-Netherlands Strategic Partnership (2026–2030): Framework and Outcomes

- **Context & Institutional Evolution:**
  - **Bilateral Elevation:** Relations upgraded to a "Strategic Partnership" (2026–2030) during PM Modi's visit to the Netherlands in May 2026.
  - **Strategic Convergence:** Focus shifted from traditional trade to high-technology partnerships ("innovation meets scale").
- **Macroeconomic & Investment Footprint:**
  - **Trade Balance:** India holds a substantial trade surplus of USD 17.393 billion.
  - **FDI Inflows:** Netherlands is India's 4th largest investor with a cumulative FDI of USD 55.6 billion.
- **Core Strategic Deliverables & Industrial Outcomes:**
  1. **Semiconductor Ecosystem:** Tata Electronics and ASML collaboration to establish a manufacturing FAB in Dholera, Gujarat.
  2. **Green Hydrogen:** Framework to expand Indian green hydrogen exports to Europe and attract infrastructure investments.
  3. **JWG on Renewable Energy:** New group targeting operational scale-up in bioenergy, battery storage, and circular feedstocks.
  4. **Critical Minerals:** Pact to secure strategic raw material supply chains and improve processing capabilities in India.
  5. **Kalpasar Project:** Technical tie-up for a massive fresh-water coastal engineering reservoir across the Gulf of Khambhat.
  6. **Customs Agreement:** Administrative pact signed to secure bilateral trade logistics and

improve regulatory enforcement.

- **Agricultural, Health, and Educational Interventions:**
  - **Agrarian Centers of Excellence (CoE):**
    - \* *Floriculture:* Established in West Tripura to promote high-value farming and export logistics.
    - \* *Dairying:* Setup at CEAH Bengaluru to enhance dairy skills, productivity, and livestock disease control.
  - **Public Health Alliance:** ICMR and RIVM partnership targeting infectious diseases, AMR, and digital health.
  - **Higher Education Partnerships:** Framework for joint degrees and faculty mobility, highlighting the Nalanda-Groningen university pact.
- **Cultural Diplomacy and Diasporic Welfare:**
  - **Chola Artefact Repatriation:** Restitution of 11th-century Chola Copper Plates mapping historical land grants in Tamil Nadu.

## 4.5 UN and It's agreements

### 4.6 Miscellaneous

#### 4.6.1 Medical and Wellness Tourism in India

- **Global Market Trends:**
  1. The global MVT market, valued at USD 115.6 billion in 2022, is projected to expand to USD 286.1 billion by 2030 at a CAGR of 10.8%.
  2. India's domestic medical tourism footprint was valued at USD 8.7 billion in 2025 and is projected to scale up to USD 16.2 billion by 2030.
  3. According to the Medical Tourism Index (MTI) 2020–21, India ranks **10th** globally in medical tourism, **12th** in wellness tourism markets, and **5th** in the Asia-Pacific region for wellness.
- **The Core Pillars of India's MVT Framework: Heal In India**
  1. **Medical Tourism (Curative Care):** Focuses on complex, high-end surgical proce-



dures, organ transplants, advanced diagnostics, and clinical interventions.

2. **Wellness Tourism (Proactive/Preventive Care):** Centered around therapeutic, preventive health via traditional systems including Yoga, Ayurveda, Naturopathy, and other AYUSH practices.

• **India's Structural and Quality Competitive Edge:**

1. **Resource Base:** Backed by 69,364 hospitals (43,486 private; 25,778 public) and 1.2 million registered doctors, fulfilling the WHO-recommended doctor-to-population ratio. English-medium clinical training removes language barriers.
2. **Accreditation Standards:** As of 2026, the National Accreditation Board for Hospitals and Healthcare Providers (NABH) has accredited over 1,299 hospitals across 600+ safety parameters. NABH is affiliated with the International Society for Quality in Healthcare (ISQua).
3. **Cost-Effective Medical Treatment:** India's medical tourism appeal is significantly strengthened by its cost competitiveness, offering high-quality treatment at a substantially lower cost compared to many developed countries while maintaining comparable clinical standards.

4. **Strengthening AYUSH-led Medical Value Travel:**

- **Unique Wellness Advantage:** India leverages its traditional systems of medicine—Ayurveda, Yoga, Naturopathy, Unani, Siddha, and Homeopathy (AYUSH)—promoting them as integral components of holistic health and preventive care.
- **AYUSH Visa Facilitation:** To streamline access, a dedicated AYUSH Visa category was introduced on July 27, 2023, allowing foreign nationals and their attendants to travel specifically for recognized traditional treatments.

- **Quality Standards:** To enhance credibility and standardization, the Bureau of Indian Standards (BIS) has officially adopted the international standard ISO 22525 for medical wellness tourism services.
- **Insurance Coverage:** Under IRDAI regulations, insurance coverage has expanded to AYUSH therapies, with around 27 insurance companies currently offering more than 140 policy products.
- **Global Engagement & Capacity Building:** MVT remains a central theme across major forums, including the "Global Synergy in AYUSH" Summits in Mumbai (2024) and Chennai (May 2025). Workforce development is supported by the AYUSH sub-council under the Health Sector Skill Council.
- **Global Outreach:** Outreach continues to expand through platforms like the WHO Global Traditional Medicine Summit, the Know India Programme, and designated showcases during the Maha Kumbh Mela 2025.

• **Budgetary and Strategic Initiatives (Union Budget 2026–27):**

1. **Regional Medical Hubs:** Plan to set up **five** Regional Medical Hubs via Public-Private Partnership (PPP) to establish integrated complexes housing healthcare, research, AYUSH facilities, and MVT Facilitation Centres under one roof.
2. **Traditional Medicine Infrastructure:** Announcement of **three** new All India Institutes of Ayurveda. Simultaneously, the WHO Global Traditional Medicine Centre in Jamnagar is being upgraded to enhance global clinical research.

• **Policy, Governance & Digital Facilitation:**

1. **Dedicated Visa Regime:** Introduction of specific **e-AYUSH Visa** and **e-AYUSH Attendant Visa** categories (launched July 27, 2023) alongside conventional e-Medical visas



extended to 172 countries.

2. **Standardization & Insurance Integration:** Bureau of Indian Standards (BIS) adopted the international ISO 22525 standard for wellness tourism. IRDAI framework has successfully onboarded  $\approx 27$  insurance firms offering over 140 policy products covering AYUSH treatments.
3. **Governance Bodies:** The National Medical & Wellness Tourism Promotion Board (NMWTB) acts as an apex inter-ministerial coordinator under the Ministry of Tourism .
4. **Capacity Building:** A pilot project under the 2026–27 Budget targets upskilling 10,000 guides at 20 iconic locations with a 12-week course. Paramedical staffs are being formally trained in cross-cultural sensibilities and foreign languages.

## Chapter 5

### Science and Technology

#### 5.1 Space Tech

##### 5.1.1 Science Collaboration as Space Compass: NASA's Lunar South Pole Mission & Geopolitics of Space

###### *NASA Lunar Roadmap – Three Phases*

- **2026–2029:** Surface missions; scientific experiments.
- **Late 2020s–early 2030s:** “Initial operations”; potential **nuclear power installations**.
- **Post-2032: Semi-permanent human settlement** on lunar South Pole.

###### *Why Lunar South Pole?*

- Shorter shadow periods → consistent **solar energy generation** (unlike 2-week light/dark cycles elsewhere on Moon).
- **Permanently Shadowed Craters:** water ice deposits → rocket fuel, life support, habitability.
- Ice records: history of water/material movement through solar system → insights into **conditions for emergence of life**.

###### *Moon as Gateway to Deep Space*

- Staging ground for **Mars missions**.
- Study human physiological adaptation to **reduced gravity** before deep-space voyages.
- Long-duration habitation tech + resource utilisation testing.

###### *Geopolitics of Space*

- **China:** lunar rovers + sample-return missions; robust programme.
- **China–Russia: International Lunar Research Station (ILRS)** – planned within 100 km of lunar South Pole; nuclear-powered.
- **USA: Artemis Programme** – led by NASA; competes with ILRS.
- Unlike Cold War's symbolic space race → current competition = **early access to strategically**

**significant extraterrestrial regions and resources.**

###### *Private Sector Role*

- **Blue Origin** (Jeff Bezos): heavy-lift rockets, lunar landers, early infrastructure for NASA's lunar base.
- **SpaceX** (Elon Musk): central partner in broader US space ambitions.
- Reflects growing importance of **public-private partnerships** in space exploration.

###### *Space Governance: Challenges*

- Cold War-era legal framework (Outer Space Treaty, 1967) **ill-equipped** for current dimensions.
- Risks: competing legal regimes; fragmentation of scientific cooperation; resource/territorial disputes.
- **Artemis Accords** (US-led) vs. **ILRS framework** (China-Russia) → parallel governance architectures.

###### *Way Forward*

- Updated **multilateral norms** for peaceful, sustainable outer space use.
- **Science, not geopolitics**, must drive space exploration.
- International collaboration essential – shared challenges (climate change, tech development) demand cooperative frameworks.

#### 5.2 Defence Tech

#### 5.3 IT and communication Tech

#### 5.4 Health

#### 5.5 Energy

#### 5.6 Bio Tech

#### 5.7 Nuclear Tech

#### 5.8 Miscellaneous

5.8.1 Major Chemistry Breakthrough: Carbon-Free Sandwich Molecule

- **The Breakthrough:** Researchers at **IIT Madras** and the **Indian Institute of Science (IISc), Bengaluru**, have achieved a major chemistry breakthrough by synthesizing a completely **carbon-free** molecule that replicates the unique sandwich structure of the famous carbon-containing **ferrocene** molecule.

Comparative Analysis: Ferrocene vs. The New Molecule

| Traditional Ferrocene                         | New Carbon-Free Molecule                                 |
|-----------------------------------------------|----------------------------------------------------------|
| Iron (Fe) atom at center                      | Osmium (Os) atom at center                               |
| Flat carbon-ringed molecules                  | Boron-based flanking rings                               |
| High carbon reliance                          | Completely <b>Carbon-free</b>                            |
| Standard organometallic bond                  | Exceptionally strong bonding                             |
| Used in medicines, batteries, and electronics | Potential for engineering novel, highly robust materials |

## Chapter 6

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### Internal Security

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## Chapter 7

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### Miscellaneous

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